

MongoDB Flow: Natural Language to Trusted Results

Natural Language -> Query Generation -> Guardrails -> Result

1) Natural Language Input

A user asks a business question in plain English. The agent extracts intent, entities, time windows, and required metrics.

Example user prompt

```
"Show Scope 1 emissions for semiconductor companies in 2024, sorted by highest emissions and limited to 25 records."
```

2) Query Generation

- The agent maps business phrases to the approved collection schema and legal MongoDB operators.
- Output is deterministic and machine-checkable before any database execution.

Generated MQL plan (pre-guardrails)

```
{
  "collection": "esg_emissions",
  "query": {"company_sector": "Semiconductors", "reporting_year": 2024},
  "projection": {"company_ticker": 1, "scoped_emissions": 1, "_id": 0},
  "sort": {"scoped_emissions": -1},
  "limit": 25
}
```

3) Guardrails

- Field allowlist: blocks unknown or policy-restricted fields.
- Index coverage: requires indexed filter paths to avoid collection scans.
- Safety injection: enforces maxTimeMS and hard result limits.

Blocked response shape

```
{
  "success": false,
  "guardrail": "index_coverage",
  "error": "No indexed filter field found",
  "details": {"required": ["company_ticker", "reporting_year"]}
}
```

4) Trusted Result

- Only validated queries execute against MongoDB.
- Results include clear user-facing output plus traceable guardrail context.

Outcome: faster insight with governance, reliability, and predictable performance.